



CS50

C++ Strings



String

C++ has in its definition a way to represent sequence of characters as an object of class. This class is called `std::string`. String class stores the characters as a sequence of bytes with a functionality of allowing access to single byte character.

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main() {
6
7      string str = "Hello, World!";
8      cout << "String: " << str << endl;
9
10     return 0;
11 }
```

ASCII

Just as numbers are binary patterns of ones and zeros, letters are represented using ones and zeros.

If you received a text message, the binary under that message might represent the numbers 72, 73, and 33. Mapping these out to ASCII, your message would look as follows:

H

I

!

72

73

33

ASCII

0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	\$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z

NUL character \0

a string is an array of characters that begins with the first character and ends with a special character called a NUL character:

H s[0]	I s[1]	! s[2]	\0 s[3]				

NUL character \0

H	I	!	\0
s[0]	s[1]	s[2]	s[3]

72	73	33	0				
s[0]	s[1]	s[2]	s[3]				

Think With ME

Given a string `s` print it's length.

Input:

ITI

Output

3

Input:

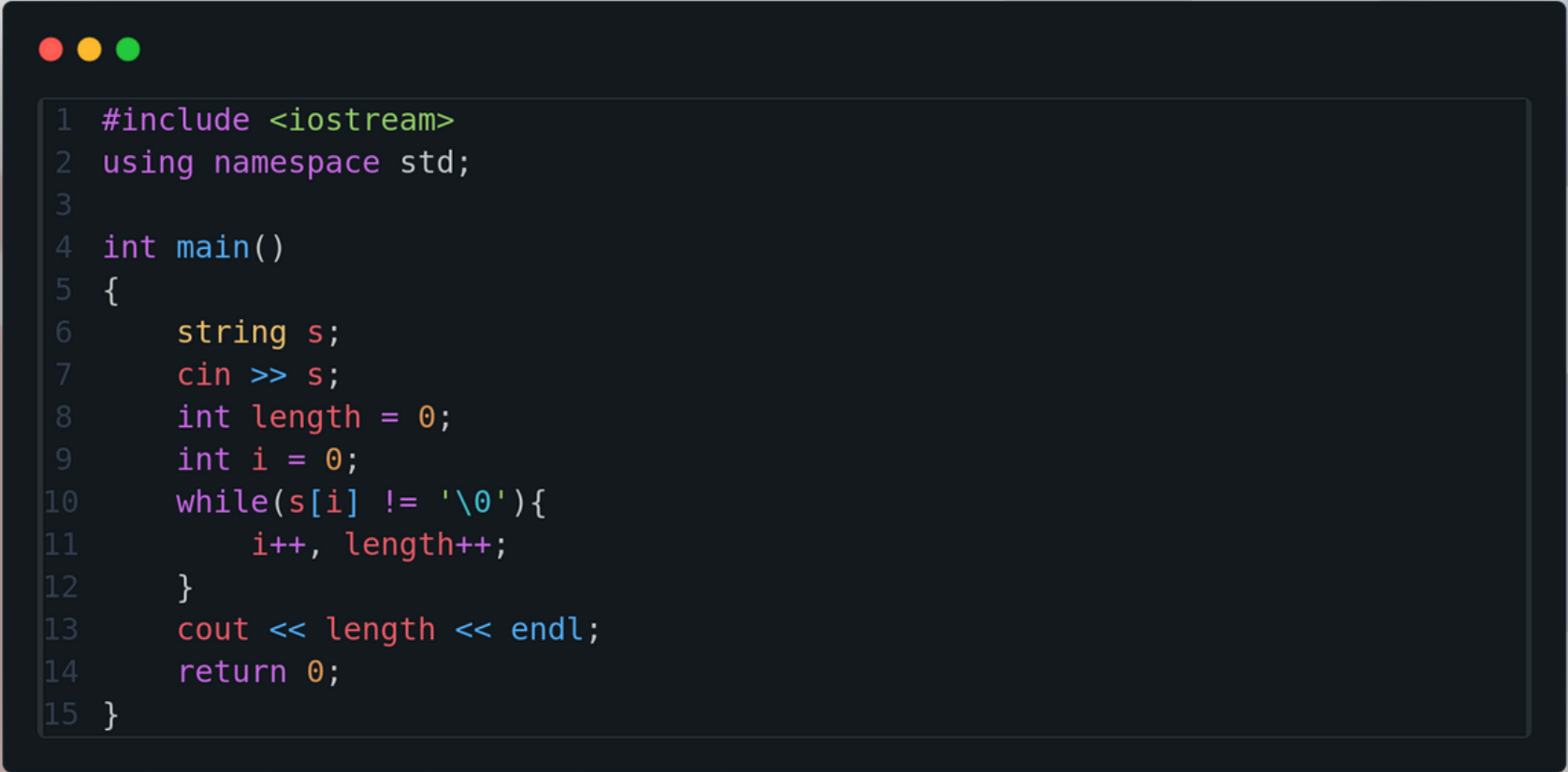
CS50

Output

4

Think With ME

Solution:



```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      string s;
7      cin >> s;
8      int length = 0;
9      int i = 0;
10     while(s[i] != '\0'){
11         i++, length++;
12     }
13     cout << length << endl;
14     return 0;
15 }
```


length & size



```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 int main() {
6     string str = "Programming";
7
8     cout << "Length: " << str.length() << endl;
9     cout << "Size: " << str.size() << endl;
10    return 0;
11 }
```

Concatenation

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main() {
6      string first = "Hello";
7      string last = "World";
8
9      string full = first + " " + last; // Using '+'
10     first += " Everyone"; // Using '+='
11
12     cout << "Concatenated: " << full << endl;
13     cout << "Updated first: " << first << endl;
14
15     return 0;
16 }
17
```

Accessing Characters

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 int main() {
6     string word = "Hello";
7
8     cout << "First Character: " << word.at(0) << endl;
9     cout << "Last Character: " << word[word.length() - 1] << endl;
10
11     return 0;
12 }
```

Substring Extraction

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4  int main() {
5      string text = "Hello, World!";
6
7      string part = text.substr(0, 5); // "Hello"
8
9      cout << "Substring: " << part << endl;
10
11     return 0;
12 }
```

Finding Substrings

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4  int main() {
5      string sentence = "I love programming";
6
7      int position = sentence.find("love");
8
9      if (position != -1)
10         cout << "Found at index: " << position << endl;
11     else
12         cout << "Not found" << endl;
13
14     return 0;
15 }
```

Replacing a Part of the String

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 int main() {
5     string str = "I hate coding!";
6     str.replace(2, 4, "love"); // Replace "hate" with "love"
7     cout << "Modified String: " << str << endl;
8
9     return 0;
10 }
```

Erasing and Clearing Strings

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 int main() {
5     string str = "Hello, World!";
6
7     str.erase(5, 7); // Remove ", World"
8
9     cout << "After erase: " << str << endl;
10
11    str.clear(); // Empty the string
12    cout << "After clear: " << str << endl;
13
14    return 0;
15 }
```

getline



```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 int main() {
5
6     string fullName;
7     cout << "Enter your full name: ";
8     getline(cin, fullName);
9     cout << "Hello, " << fullName << "!" << endl;
10    return 0;
11 }
```




Support
Team

